Next step (i): the checkpoint's figures on TinyStories-30M and Shakespeare

Each figure of the checkpoint (TinyStories-10M, first) is redone for TinyStories-30M and character-level Shakespeare, AdamW and Muon only, with the checkpoint's protocol: one initialization, five shared data orders, each optimizer at the best of five learning rates (TS-30M: AdamW $2 \cdot 10^{-3}$, Muon 0.01; Shakespeare: AdamW $3 \cdot 10^{-4}$, Muon 0.04, picked by the lowest validation loss along training, as the runs overfit). Every step (checkpoints, marked steps, fork steps) is scaled by the training length: $\times 2$ for TS-30M (20,000 steps), $\times 0.15$ for Shakespeare (1500 steps).

1 Figure 1: how the trajectories differ

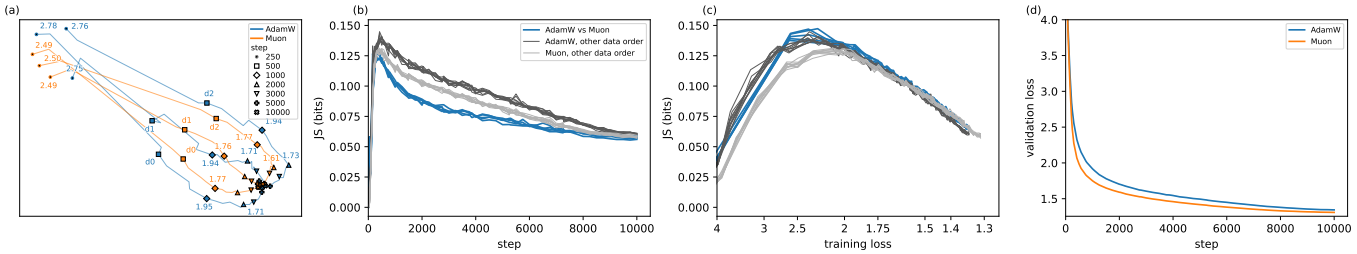


Figure 1a: TinyStories-10M (the checkpoint's Figure 1). (a) Checkpoints from step 250 on, mapped to 2-D by densMAP; labels: training loss and data order. (b) JS between runs at equal step. (c) The same at equal training loss. (d) Validation loss.

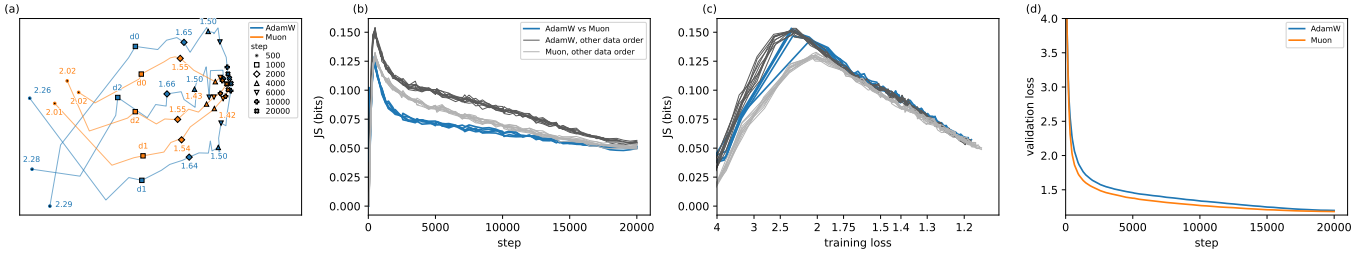


Figure 1b: TinyStories-30M, as **1a** with the steps doubled: checkpoints from step 500 on.

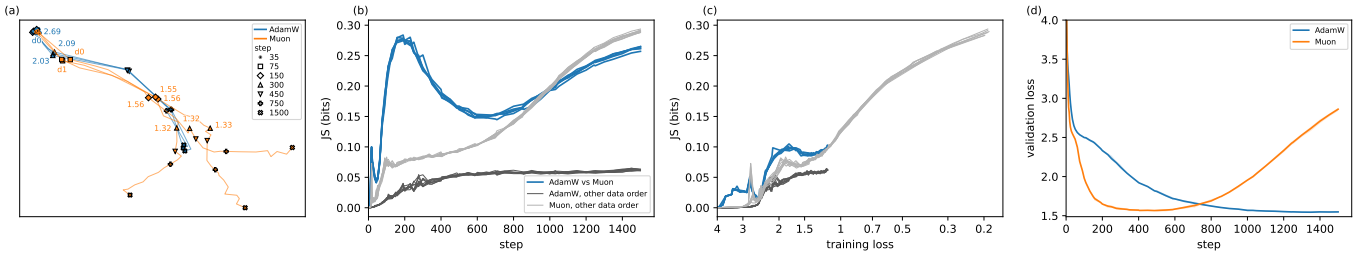


Figure 1c: Shakespeare, as **1a** with the steps $\times 0.15$: checkpoints from step 35 on. Muon overfits (d), and its training loss goes below 1, so (c) uses a log scale in the loss instead of in the loss minus 1.

2 Figure 2: what differs along the way

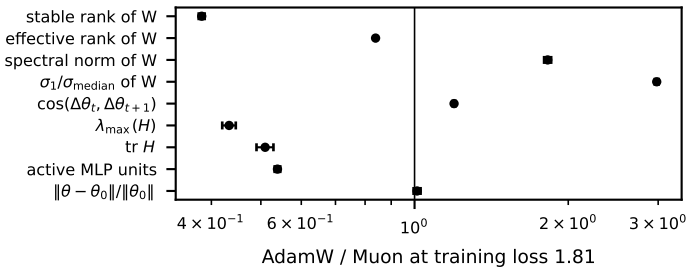


Figure 2a: TinyStories-10M (the checkpoint's Figure 2): ratio AdamW/Muon at training loss 1.81; three data orders each; bars: range across data orders.

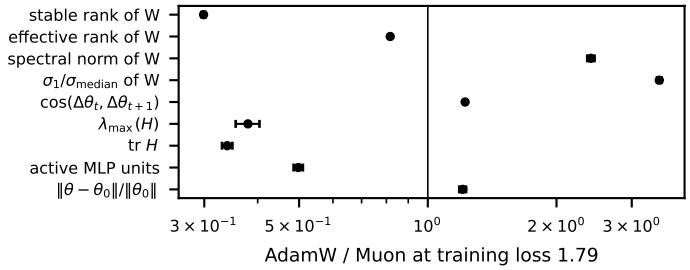


Figure 2b: TinyStories-30M, as **2a**, at training loss 1.79 (the level of the loss grid nearest 1.81).

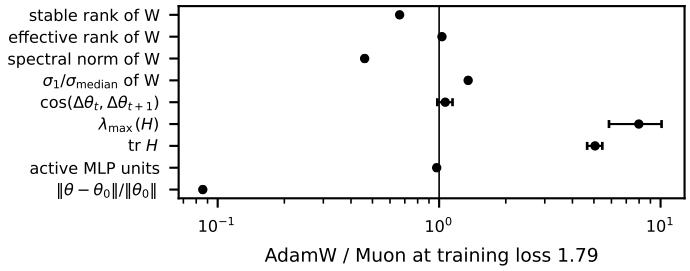


Figure 2c: Shakespeare, as **2a**, at training loss 1.79 (the level of the loss grid nearest 1.81).

3 Figure 3: swaps

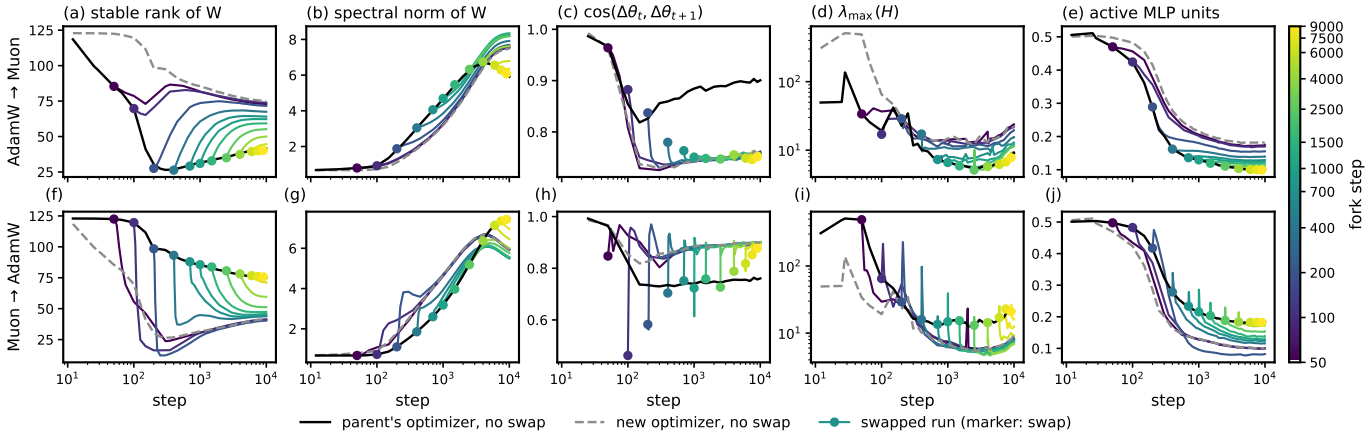


Figure 3a: TinyStories-10M: the checkpoint's Figure 3 with swaps at 6000, 7500 and 9000 added. Top: AdamW to Muon; bottom: Muon to AdamW. Black: the parent's optimizer without a swap; grey dashed: the new optimizer's own run; colour: swapped runs, by fork step (dot: the swap).

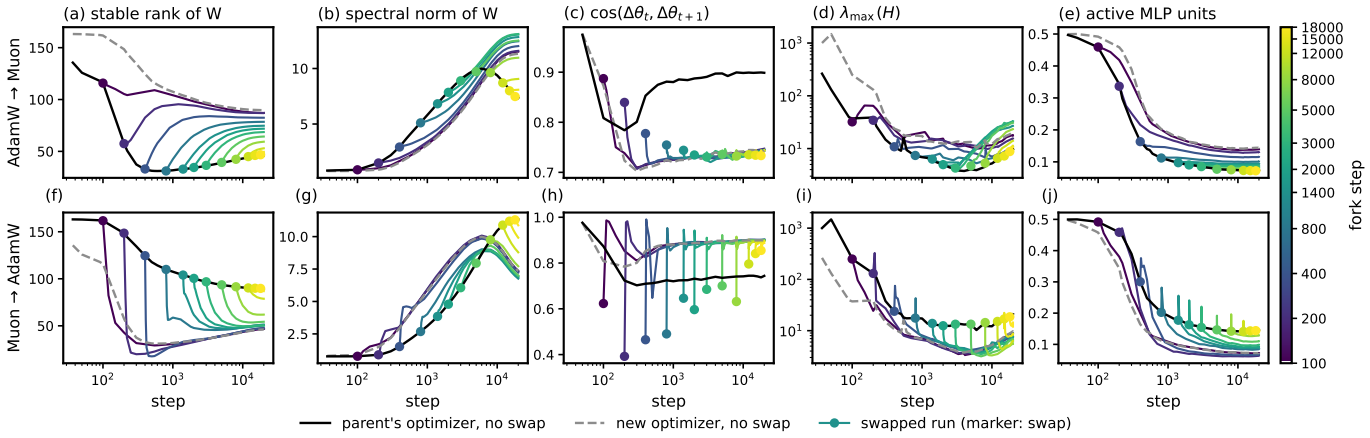


Figure 3b: TinyStories-30M, as **3a** with the fork steps doubled (100 to 18,000).

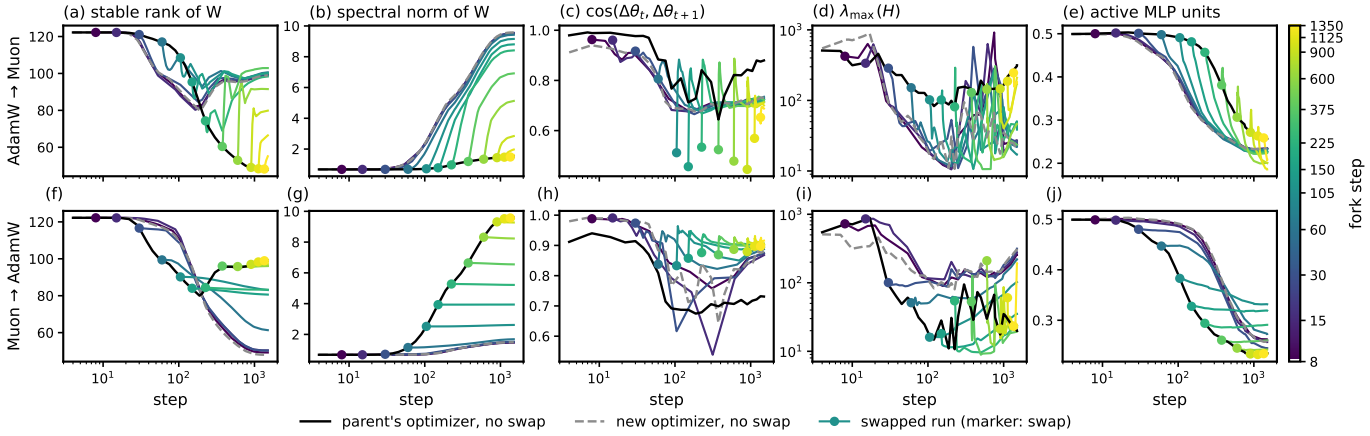


Figure 3c: Shakespeare, as **3a** with the fork steps ×0.15 (8 to 1350).